



AUTOXR: REAL-TIME AR/VR VEHICLE CUSTOMIZATION

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Problem Statement

Market Needs & Value Proposition

- Vehicle buyers struggle to visualize colors, parts, and accessories accurately before purchase.
- This leads to uncertainty, long decision cycles, and costly rework for dealers and custom shops.
- AutoXR solves this by projecting full-scale, true-to-life AR customization directly onto the actual vehicle.
- Survey (N=40):
 - **72.5%** prefer AR over standard previews
 - **65%** spend **3+ hours** customizing
 - **82.5%** choose price ranges **< \$2,000**
 - AutoXR creates value through realism, faster decisions, and fewer ordering mistakes.

AR Training for Automotive Mechanics

Conjoint Insights and Value Proposition

- **Product:** AutoXR is an XR training and demo solution for automotive customers.
- **Experience:** Uses immersive XR to show vehicle features and guide complex tasks.
- **Goal:**
 - Help users understand advanced features faster.
 - Reduce confusion during the sales process and after delivery.
- **Study:**
 - Online choice-based **conjoint survey**.
 - Attributes: Plans, XR realism, where you can use it, saved configurations, collaboration features, monthly price.
 - We use the results to define **target segments** and a **pricing strategy**.



Insights, Methods & Pricing Logic

Data, Methods & EVC Market Anchor

Plans (Importance Score)	XR Realism (Importance Score)	Where you can use it (Importance Score)	Saved configurations (Importance Score)	Collaboration features (Importance Score)	Monthly price (Importance Score)
0.118663548	0.044594909	0.190024727	0.077225041	0.091433885	0.47805789
0.093898516	0.192345721	0.041596359	0.117605628	0.203665599	0.350888176
0.1069836	0.078506837	0.183578219	0.076824547	0.071221046	0.482885751
0.085958591	0.136475105	0.131155103	0.100244426	0.085945371	0.460221404
0.134883879	0.161686129	0.05226401	0.081575833	0.207149756	0.362440393
0.066179747	0.204851681	0.021645655	0.143855411	0.314092579	0.249374926
0.109032527	0.04835253	0.175323631	0.069808666	0.096389483	0.501093162

Data & Methods:

- 40 survey responses on customization habits, time spent, AR usefulness, and price perception.
- Analysis combined descriptive statistics, willingness-to-pay indicators, and competitor benchmarks.

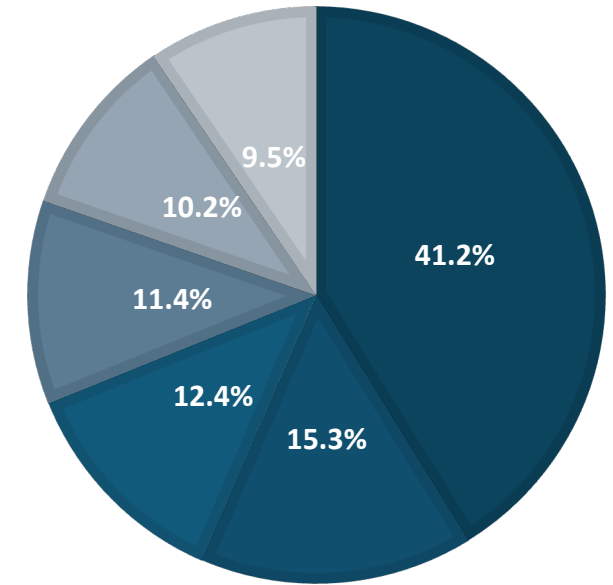
EVC Market Anchor:

- Competitor pricing:
 - Unreal: **\$1,850/year**
 - Unity Forma: **Enterprise (5–6 figures)**
 - ZeroLight: **\$100k+/year**
- Survey anchors:
 - **82.5%** expect costs < \$2k
 - Supports AutoXR’s target pricing: **\$499–\$1,499/month** for dealers.

What Drives Choice Preferences - Attribute Importance

- Our conjoint results show the following **importance scores**:
 - Monthly price: 41.2%
 - Collaboration features: 15.3%
 - XR Realism (content quality): 12.4%
 - Where you can use it (showroom vs showroom + home): 11.4%
 - Plans (feature bundle): 10.2%
 - Saved configurations: 9.5%
- **Interpretation:**
 - Price is the single largest driver of choice, which fits a recurring monthly subscription.
 - Non-price features still matter a lot: collaboration + XR realism + access location together account for almost 40% of decisions.
 - Customers value useful collaboration tools and high-quality XR realism, not just a cheap plan.

ATTRIBUTE IMPORTANCE SPLIT



- Monthly price
- Collaboration features
- XR realism (content quality)
- where you can use it (showroom vs showroom+home)
- plans (feature bundle)
- saved configurations

Utilities and Main Drivers

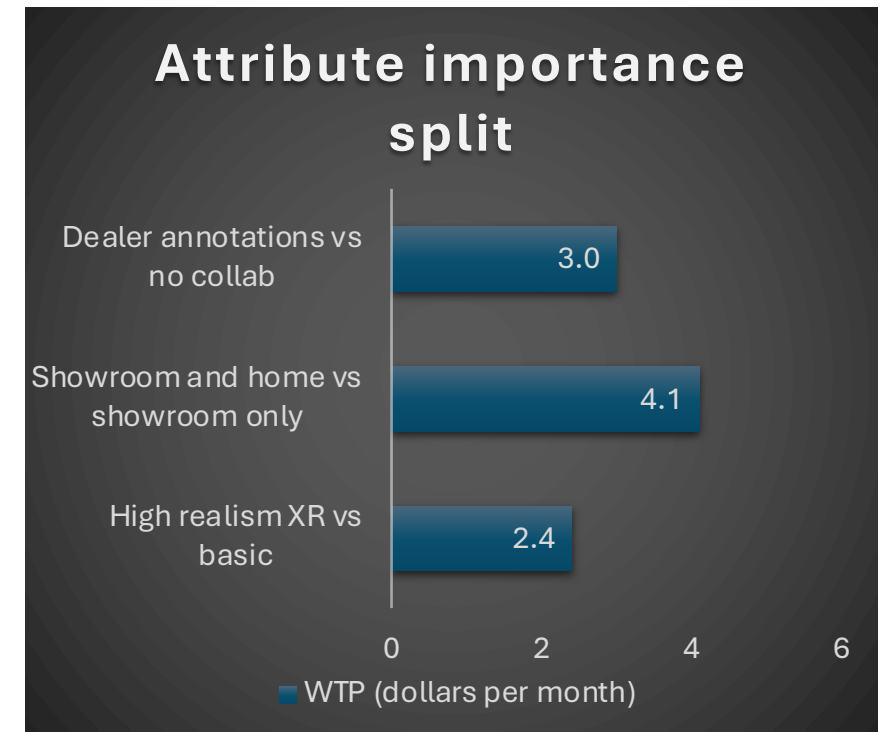
Part Worth Utilities – Best vs
Worst Profiles

- **XR realism (content):**
 - Level 3 (high detail + reflections): **+37.42 utils**,
 - Level 2 (lighting only): **-36.88 utils** or
 - Level 1 (basic overlay): **-0.54 utils**
 - → Only **full realism (Level 3)** is clearly preferred.
- **Where you can use it (Access):**
 - Showroom + at home: **+32.24 utils** or
 - Showroom only: **-32.24 utils** → Strong penalty for **showroom-only** experiences.
- **Collaboration:**
 - Dealer annotations: **+20.3 utils** Friends/family sharing: **+6.7 utils** or None: **-27.0 utils** → Customers want **professional guidance**, not no-collab.
- **Price:** \$5/month: **+135.8 utils**, \$10/month: **-38.2 utils** or \$20/month: **-97.6 utils** → Clear **price sensitivity** across the range.
- **Overall:**
 - **Best profile:** Level 3 realism, showroom + home, dealer annotations, lower price.
 - **Worst profile:** Mid/low realism, showroom-only, no collaboration, high price.

Willingness to Pay for Key Features

Willingness to Pay – Using the Price Attribute

- From the **price utilities** (5 dollars vs 20 dollars per month), the model implies:
 - One util is worth about **\$0.06 per month**
- **Extra willingness to pay per month compared to a basic option:**
 - **High realism XR (Level 3 instead of Level 1)**
 - Utility difference is about 38 utils. This is worth about **\$2.4 more per month**
 - **Use AutoXR in the showroom and at home (instead of showroom only)**
 - Utility difference is about 64.5 utils. This is worth about **\$4.1 more per month**
 - **Dealer annotations (instead of no collaboration)**
 - Utility difference is about 47.3 utils. This is worth about **\$3.0 more per month**
- **Interpretation (link to lecture concepts):**
 - These numbers show the **monetary value of each feature** on the demand curve.
 - Customers are willing to pay several dollars more per month for:
 - Better content, more usage occasions, better guidance.
 - This supports a **two-tier strategy**: a basic plan and a premium plan that captures this extra value.



Take-up vs Price

Simulation setup:

- Premium price varies; Basic is fixed at \$10/month
- Premium bundle includes Plan B, Level 3 realism, showroom + home, 10 saves, dealer annotations

Key patterns from the simulation:

- Premium adoption is strong in the \$14–\$15/month range
 - Take-up remains around 50–58% in this zone
- As price increases toward \$20/month,
 - None share grows sharply
 - Many users drop out rather than accept a high premium price
- Basic plan stays weak because it combines low realism, showroom-only access, and no collaboration, all of which have negative utilities

Implication:

- The \$14–\$15/month range is a good starting point for a Premium tier: still attractive, without competing purely on being the cheapest option.

Recommended Configuration & Price

Recommended Premium configuration (v1):

Plan: Plan B
XR realism: Level 3 (high detail + reflections)
Where to use: Showroom and home
Saved configurations: Up to 10 setups
Collaboration: Dealer annotations

Why this configuration:

Aligns with highest utilities on realism, access, and collaboration
Matches survey attitudes: realism and confidence (Q5, Q6, Q8) and desire to use AutoXR (Q11)

Recommended pricing:

Basic: \$10/month (low realism, showroom-only, minimal collaboration)
Premium: \$14–\$15/month
≈ 70–75% of calculated EVC (~\$19.6/month)
Supported by simulation take-up and willingness-to-pay
Dealer package: AutoXR Pro at \$150/location/month with ROI justification (upsell, faster explanations, fewer post-sale questions)

Overall Conclusion



1. Product fit:

AutoXR's value lies in high realism, showroom + home access, and guided collaboration (dealer annotations)



2. Customer segment:

Best near-term fit: OEM dealerships and larger independent shops serving buyers who already customize online or in showrooms



3. Pricing strategy:

Use a tiered structure rather than a single flat price
Position Premium in the \$14–\$15/month range (per end-user equivalent), anchored by EVC and simulations

For dealers, a \$150/location/month subscription is perceived as close to “about right” on average



4. Adoption risk:

Comfort (motion sickness) and privacy are key concerns that must be actively managed

Any Questions?

Thank you

